

Exact colour sums for multi-gluon amplitudes: direct, multiplet and Fourier methods

Rikkert Frederix,^{1,*} Valentin Hirschi,^{2,†} and Malin Sjödhall^{1,‡}

¹*Department of Physics, Lund University, Box 118, 221 00 Lund, Sweden*

²*Institute for Theoretical Physics, University of Bern, Sidlerstrasse 5, 3012 Bern, Switzerland*

We compare exact $SU(3)$ colour sums for fixed-helicity, tree-level all-gluon matrix elements. In the trace and adjoint decompositions, a direct colour contraction couples every pair of partial amplitudes, so its evaluation time grows quadratically with the factorially growing number of gluon orderings. An orthonormal multiplet basis removes this final double sum but requires a representation-labelled amplitude recursion. Alternatively, a fast Fourier transform (FFT) on the symmetric group retains the trace or adjoint partial amplitudes and reorganizes the exact colour sum into smaller contractions labelled by Young diagrams. We compare CPU time after initialization and maximum memory use for four through eleven gluons. At eleven gluons, one matrix-element evaluation takes 19.8 s in the multiplet basis, 3.08 s with the trace FFT, and 0.814 s with the adjoint FFT; the direct trace and adjoint calculations are estimated to take 2.41×10^5 s and 1.69×10^3 s. The adjoint FFT is the fastest method from six through eleven gluons.

I. INTRODUCTION

Colour-ordered Berends–Giele recursion efficiently computes tree-level gluon partial amplitudes [1]. Computing an exact colour-summed squared matrix element also requires evaluating all interference terms among the associated colour structures. At fixed external momenta and helicities, let $\mathcal{M}$ denote the full amplitude. The index α labels the chosen colour tensors C_α , and A_α is the corresponding complex kinematic coefficient:

$$\mathcal{M} = \sum_{\alpha} C_{\alpha} A_{\alpha}. \quad (1)$$

The sum over all external colours is then

$$\begin{aligned} \mathcal{B} &\equiv \sum_{\text{colours}} |\mathcal{M}|^2 = \sum_{\alpha, \beta} A_{\alpha}^* G_{\alpha\beta} A_{\beta}, \\ G_{\alpha\beta} &\equiv \sum_{\text{colours}} C_{\alpha}^* C_{\beta}. \end{aligned} \quad (2)$$

The star denotes complex conjugation, the vertical bars denote the complex modulus, and α and β both label the colour tensors in Eq. (1). The quantity $\mathcal{B}$ is the colour-summed squared matrix element, while G is the colour-overlap matrix with entries $G_{\alpha\beta}$. The structure of G , and hence the work required to evaluate Eq. (2), depends on the chosen colour decomposition.

The trace decomposition [2, 3] and the adjoint basis of Del Duca, Dixon and Maltoni [4] express the amplitude in terms of familiar colour-ordered partial amplitudes. Their colour tensors are not orthogonal, so the off-diagonal terms in Eq. (2) must be retained. In both cases, the number of orderings grows factorially with the gluon

multiplicity, making direct evaluation of the double sum increasingly expensive.

Multiplet bases offer a different organization of colour space [5, 6]. They are constructed by coupling gluons through irreducible $SU(3)$ representations, and their colour tensors can be chosen orthonormal. The final colour sum then reduces to a sum of absolute squares. This simplification requires the recursion to track representation labels and recouple the corresponding colour states when gluons are reordered. Here we present the first numerical implementation of the off-shell multiplet recursion developed in Ref. [7]. We also use this recursion for the first time to evaluate fully colour-summed squared matrix elements at fixed momenta and helicities.

The non-orthogonal trace and adjoint decompositions nevertheless have useful structure. In either case, the overlap between two colour tensors depends only on the relative permutation of their gluon orderings. A fast Fourier transform (FFT) on the symmetric group exploits this property to replace the direct double sum with a set of smaller, independent matrix contractions. The Fourier method is exact: it retains every colour interference term and does not change the partial amplitudes. To our knowledge, this is the first application of a symmetric-group FFT to exact colour summation for QCD matrix elements.

We examine these three approaches in turn. Section II describes direct colour summation in the trace and adjoint decompositions. Section III presents the orthogonal multiplet basis and its recursion. Section IV explains the symmetric-group Fourier method for the trace and adjoint colour sums. Section V compares their CPU times per evaluation after initialization and their maximum memory use. Throughout, N denotes the total number of external gluons.

* rikkert.frederix@fysik.lu.se

† hirschi@itp.unibe.ch

‡ malin.sjodahl@fysik.lu.se

II. DIRECT COLOUR SUMS IN TRACE AND ADJOINT DECOMPOSITIONS

A. Trace decomposition

Cyclic invariance lets us fix one external gluon in the single-trace decomposition. For any positive integer q , S_q denotes the group of permutations of q labels; $\sigma \in S_q$ is one such permutation, and $\sigma(i)$ is the image of label i . Permuting the remaining $N - 1$ gluons gives

$$\mathcal{M} = \sum_{\sigma \in S_{N-1}} C_{\sigma}^{\text{tr}} A(\sigma(1), \dots, \sigma(N-1), N), \quad (3)$$

with

$$P_{\text{tr}} = (N - 1)! \quad (4)$$

partial amplitudes, where $!$ denotes factorial. Here C_{σ}^{tr} is the trace colour tensor for order σ , $A(\sigma(1), \dots, \sigma(N-1), N)$ is the corresponding partial amplitude, and P_{tr} is the number of trace orders. The trace set is overcomplete but convenient because its kinematic coefficients are the usual colour-ordered amplitudes. Reflection relates each ordering to its reverse, so the corresponding currents can be shared when evaluating partial amplitudes.

B. Adjoint basis

The adjoint decomposition places gluons 1 and N at the ends of a chain of structure constants and permutes the remaining gluons [4]:

$$\mathcal{M} = \sum_{\sigma \in S_{N-2}} C_{\sigma}^{\text{adj}} A(1, \sigma(2), \dots, \sigma(N-1), N). \quad (5)$$

There are

$$P_{\text{adj}} = (N - 2)! \quad (6)$$

partial amplitudes. Here C_{σ}^{adj} is the chain of structure constants for order σ , $A(1, \sigma(2), \dots, \sigma(N-1), N)$ is the corresponding partial amplitude, and P_{adj} is the number of adjoint orders. The adjoint set is smaller than the trace set, but its colour tensors remain non-orthogonal.

C. Direct contraction

For either decomposition, let P denote P_{tr} or P_{adj} , as appropriate. For trace, let σ , τ , and π belong to S_{N-1} ; for adjoint, let them belong to S_{N-2} , which here permutes labels $2, \dots, N - 1$. Direct evaluation of Eq. (2) contracts one pair of orders at a time. Because the colour matrix is real and symmetric, we evaluate each diagonal term once, while each off-diagonal pair contributes twice the real part of its upper-triangular term. A common relabelling of both colour tensors leaves their overlap unchanged, so

$$G_{\sigma\tau} = k(\sigma^{-1}\tau), \quad k(\pi) = k(\pi^{-1}). \quad (7)$$

Here $G_{\sigma\tau}$ is the overlap of the colour tensors for the two orders, the superscript -1 denotes permutation inversion, and $k(\pi)$ is the overlap obtained by placing the first tensor in the reference order and using π as the relative permutation. Only the P values of k need to be stored because every row of G is a permutation of this list. The contraction still considers P^2 pairs, so its time scales as $[(N - 1)!]^2$ for trace and $[(N - 2)!]^2$ for adjoint. These direct contractions provide the baseline for the comparison below.

For both decompositions, we use Berends–Giele recursion [1] to evaluate the ordered amplitudes. We compute currents shared by several complete orders once and reuse them. Our comparison therefore includes both partial-amplitude evaluation and final colour contraction, rather than the contraction alone.

III. THE MULTIPLY BASIS

Each gluon transforms in the adjoint representation of $SU(3)$. To construct a multiplet basis, we couple the first two gluons to an irreducible representation and then couple each remaining gluon in turn until the total representation is a singlet. Each basis vector is labelled by the resulting representation sequence and an extra label whenever a representation occurs more than once. With a suitable normalization, the colour tensors obey

$$\sum_{\text{colours}} C_{\alpha}^* C_{\beta} = \delta_{\alpha\beta}. \quad (8)$$

Here α and β label basis vectors, including any extra labels for repeated representations, and $\delta_{\alpha\beta}$ is the Kronecker delta. Equation (2) therefore becomes

$$\mathcal{B} = \sum_{\alpha} |A_{\alpha}|^2. \quad (9)$$

Thus, once the amplitude coefficients A_{α} are known, exact colour summation reduces to an ordinary sum of absolute squares.

We compute the amplitude coefficients in this basis using the off-shell recursion of Ref. [7]. A current is labelled by a subset of external gluons and by an allowed sequence of $SU(3)$ representations. Joining two currents requires reordering their gluons into the order defining the parent basis. We decompose this reordering into exchanges of neighbouring gluons. Each exchange acts on the representation sequences through a normalized matrix of Wigner $6j$ coefficients [8].

These coefficients depend on the normalization, phase, and multiplicity conventions chosen for the three-point $SU(3)$ vertices. We calculate these coefficients in the multiplet-basis conventions and tabulate the resulting normalized exchange matrices. We validate the matrices in three ways: by direct contractions of explicit adjoint tensors, by contractions after resolving each adjoint into a traceless fundamental–antifundamental pair, and by

recursive reduction to the elementary two-line recouplings described in Refs. [9, 10]. The resulting tabulation contains 354 exchange blocks with 5673 nonzero entries and covers every recoupling required at the multiplicities considered. Because the coefficients and exchange sequences depend only on N , we prepare them once before evaluating phase-space points.

The representations allowed in repeated adjoint products determine the number of multiplet coefficients. For $N = 4, \dots, 11$, the respective counts are 8, 32, 145, 702, 3598, 19280, 107160, and 614000. The multiplet basis is smaller than the trace set from $N = 7$ onward but larger than the adjoint basis throughout this range. Orthogonality eliminates the final double sum, whereas the representation-labelled recursion requires more work than the ordered trace and adjoint recursions. The timings below show whether eliminating the double sum offsets this additional work.

IV. SYMMETRIC-GROUP FFT FOR TRACE AND ADJOINT COLOUR SUMS

For either decomposition, let $P = m!$, where $m = N - 1$ for trace and $m = N - 2$ for adjoint. The partial amplitudes define a function $A(\sigma)$ on the permutation group S_m . By Eq. (7), each colour overlap depends only on the relative permutation. The colour sum is therefore a convolution on S_m . Specifically, for each $\tau \in S_m$, the convolution sums $A(\sigma)k(\sigma^{-1}\tau)$ over $\sigma \in S_m$; the colour sum then contracts the result with $A(\tau)^*$ and sums over τ .

On a cyclic group, a Fourier transform converts convolution into scalar multiplication. Because permutations do not generally commute, the corresponding transform on S_m instead produces matrices. Its blocks are labelled by partitions λ of m , or equivalently by Young diagrams with m boxes. We write $\lambda \vdash m$ for such a partition. If d_λ is the dimension of the irreducible representation labelled by λ , then

$$\sum_{\lambda \vdash m} d_\lambda^2 = m!. \quad (10)$$

Thus the $m!$ amplitude values are reorganized exactly into one $d_\lambda \times d_\lambda$ matrix for each Young diagram.

For a complex-valued function f on S_m , define its transformed matrix $\hat{f}_\lambda$ by

$$\hat{f}_\lambda = \sum_{\sigma \in S_m} f(\sigma) \rho_\lambda(\sigma), \quad (11)$$

where $\rho_\lambda(\sigma)$ is the real orthogonal $d_\lambda \times d_\lambda$ matrix representing the permutation σ in the irreducible representation labelled by λ . Let $\hat{A}_\lambda$ be the transform of the partial-amplitude function $A(\sigma)$, and let K_λ be the transform of the colour-overlap function $k(\sigma)$. The exact colour sum is

$$\mathcal{B} = \frac{1}{m!} \sum_{\lambda \vdash m} d_\lambda \text{Tr}(\hat{A}_\lambda^\dagger \hat{A}_\lambda K_\lambda). \quad (12)$$

The dagger denotes Hermitian conjugation. Equation (12) couples only matrices associated with the same Young diagram, rather than all pairs of permutations.

Computing every transformed matrix entry directly from Eq. (11) would still be expensive. Fast transforms on finite groups instead reuse partial results [11]. For the symmetric group, we evaluate the transform recursively along the subgroup chain

$$S_1 \subset S_2 \subset \dots \subset S_m, \quad (13)$$

where S_i is the permutation group on i labels for $i = 1, \dots, m$. For $i = 1, \dots, m - 1$, the inclusion $S_i \subset S_{i+1}$ leaves label $i + 1$ unchanged. Partial results at one level are reused at the next. Along this chain, the matrices representing exchanges of adjacent gluons split into one- and two-dimensional blocks, reducing the work at each step [12, 13].

For each N , we calculate the K_λ matrices once and reuse them. Collectively, these matrices contain $P = \sum_{\lambda \vdash m} d_\lambda^2$ colour entries. At each new phase-space point and helicity assignment, we transform the partial amplitudes and evaluate the contractions in Eq. (12). The transform does not make either set of colour tensors orthogonal or reduce the number of partial amplitudes; it reorganizes the exact colour sum to avoid contracting P^2 pairs directly.

In our trace calculation, we obtain the colour overlaps by closing fundamental colour lines in $U(3)$. Although individual $U(3)$ overlaps need not equal their $SU(3)$ counterparts, the $U(1)$ component decouples from a tree-level pure-gluon amplitude. Contracting these $U(3)$ overlaps with the physical partial amplitudes therefore gives the same final $SU(3)$ colour sum. For the adjoint decomposition, we write the chains of structure constants as nested commutators. Their $U(1)$ components vanish because the identity generator commutes with every generator. The trace and adjoint cases use different values of m and different overlap functions k , but the same symmetric-group transform.

V. NUMERICAL COMPARISON

A. Setup and checks

We study $gg \rightarrow (N - 2)g$, summing the colours of all incoming and outgoing gluons while holding their helicities fixed.

For each $N = 4, \dots, 11$, we apply all five methods to the same phase-space point and the same helicity assignment, chosen to give a nonzero matrix element. We use the general Berends–Giele recursion in every case, even when an analytic expression is available, so special-case formulas do not affect the comparison. At fixed N , the recursion performs the same operations for every helicity assignment not excluded by the tree-level selection rule.

We compiled the five implementations with GNU Fortran 13.3 at optimization level -O3 and ran each on a

TABLE I. CPU time for one evaluation of a fixed-helicity, colour-summed squared matrix element after initialization. Initialization and the first evaluation are excluded. Direct denotes colour-overlap-matrix contraction without a Fourier transform, in either the trace decomposition or the adjoint basis. FFT denotes the exact symmetric-group Fourier method. Asterisks mark estimates, not measurements.

N	direct colour sum		orthogonal basis multiplet	Fourier colour sum	
	trace	adjoint		trace FFT	adjoint FFT
4	0.729 μ s	0.543 μ s	0.923 μ s	0.968 μ s	0.645 μ s
5	6.23 μ s	2.44 μ s	6.38 μ s	4.41 μ s	2.62 μ s
6	138 μ s	15.3 μ s	62.3 μ s	26.2 μ s	13.3 μ s
7	6.16 ms	0.196 ms	0.718 ms	0.186 ms	0.0832 ms
8	0.381 s	5.78 ms	8.26 ms	1.62 ms	0.620 ms
9	29.7 s	0.303 s	0.104 s	0.0155 s	0.005 23 s
10	2.41×10^3 s*	20.9 s	1.49 s	0.215 s	0.0589 s
11	2.41×10^5 s*	1.69×10^3 s*	19.8 s	3.08 s	0.814 s

*Estimated from the last measured direct result, assuming that the time is proportional to the square of the number of colour orders, with $P_{\text{tr}} = (N - 1)!$ and $P_{\text{adj}} = (N - 2)!$.

single core of an Intel Core i7-8700K. Timing excludes initialization and the first evaluation, during which any remaining N -dependent data are prepared. The reported CPU time per subsequent matrix-element evaluation is the median of ten measurements.

For each method, we measured maximum physical memory in a separate process; the values include initialization and all evaluations.

For every tested configuration, all completed calculations agree on the colour-summed squared matrix element. The largest pairwise relative difference between nonzero results is 8.1×10^{-12} , at ten gluons. For helicity configurations forbidden by the tree-level selection rule, all five methods return exactly zero. Extrapolated times are excluded from these checks.

B. Evaluation time after initialization

Table I and Fig. 1 summarize the timing results. Direct adjoint contraction is fastest at four and five gluons; adjoint with the Fourier transform is fastest from six through eleven gluons. The direct double sum becomes costly by eight gluons: direct trace takes 0.381 s, whereas trace with the transform takes 1.62 ms. At nine gluons, the corresponding times are 29.7 s and 15.5 ms, a factor of about 1900.

The multiplet method avoids the direct double sum. It is faster than direct trace at six gluons and faster than direct adjoint at nine gluons. At eleven gluons, the multiplet evaluation takes 19.8 s, versus an estimated 1.69×10^3 s for direct adjoint and measured times of 3.08 s and 0.814 s for trace and adjoint with the transform, respectively. The adjoint Fourier method is therefore fastest at this multiplicity.

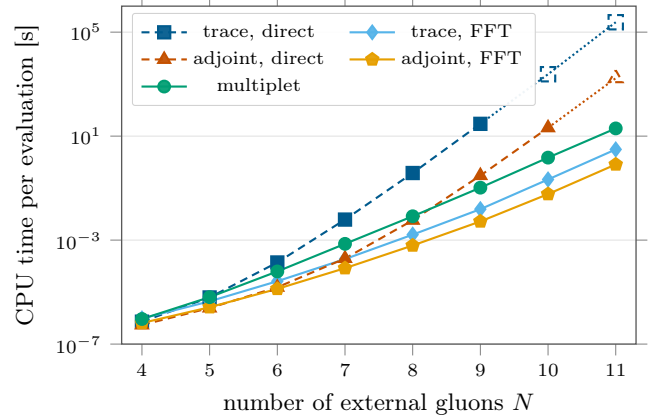


FIG. 1. CPU time to evaluate one fixed-helicity, colour-summed squared matrix element after initialization. Dashed lines show direct trace and adjoint colour sums; solid blue and orange lines show their exact Fourier counterparts; the solid green line shows the multiplet method. Dotted extensions and open symbols indicate the three estimates identified by asterisks in Table I, while filled symbols indicate measurements.

C. Memory use

The following table gives maximum memory use at the four largest multiplicities. A dash indicates that the corresponding run did not finish; no memory value is estimated.

method	$N = 8$	$N = 9$	$N = 10$	$N = 11$
trace, direct	4.24 MiB	14.0 MiB	—	—
adjoint, direct	3.34 MiB	7.66 MiB	43.9 MiB	—
multiplet	10.1 MiB	55.4 MiB	425 MiB	3.96 GiB
trace, FFT	4.43 MiB	14.8 MiB	111 MiB	1.09 GiB
adjoint, FFT	3.54 MiB	7.86 MiB	44.6 MiB	398 MiB

At every multiplicity with a direct result, the direct and Fourier variants use similar amounts of memory. At nine gluons, direct trace uses 14.0 MiB and trace with the

transform uses 14.8 MiB; the adjoint values are 7.66 MiB and 7.86 MiB. Our direct implementations store only the P distinct overlaps from Eq. (7), not the full $P \times P$ matrix. Within each decomposition, the direct and Fourier methods therefore differ far more in evaluation time than in stored colour data.

At eleven gluons, memory use is 3.96 GiB for the multiplet method, 1.09 GiB for trace with the transform, and 398 MiB for adjoint with the transform. For each of the three methods with results at both multiplicities, memory use grows by about an order of magnitude from ten to eleven gluons, suggesting that memory may become an important limitation beyond the range studied here.

VI. CONCLUSIONS

We have compared three exact approaches to evaluating fixed-helicity, colour-summed squared matrix elements for tree-level all-gluon scattering. Although our direct trace and adjoint implementations each store only one row of colour overlaps, both contractions couple every pair of partial amplitudes and therefore scale quadratically with the factorially growing number of amplitudes.

The multiplet-basis calculation reduces the final colour sum to a sum of absolute squares and substantially outperforms the direct contractions at high multiplicity, but the symmetric-group Fourier method is faster still. It retains the trace or adjoint partial amplitudes while exploiting the relative-permutation dependence of their colour overlaps to replace the direct double sum with exact contractions labelled by Young diagrams.

At eleven gluons, the CPU times per evaluation after initialization are 3.08 s for trace with the transform and 0.814 s for adjoint with the transform, compared with 19.8 s in the multiplet basis. Both Fourier methods also use less memory at this multiplicity, despite the multiplet basis's simpler final colour sum. The direct trace and adjoint contractions are estimated to require 2.41×10^5 s and 1.69×10^3 s, respectively. The adjoint Fourier method uses the smaller $(N - 2)!$ set of partial amplitudes; it is fastest from six through eleven gluons and has the smallest recorded memory use at eleven gluons.

Processes with external quarks are a natural next application. The pure-gluon adjoint chain cannot be carried over unchanged because quarks and antiquarks carry fundamental and antifundamental colour. For one quark-antiquark pair, the corresponding colour decomposition uses an open chain of fundamental generators, with the gluons permuted between the quark and antiquark. For several pairs, the established generalization also tracks the placement and nesting of the quark lines [14, 15]. Multiplet bases can accommodate external quarks, antiquarks, and gluons while remaining orthogonal, but extending the recursion requires representation paths and recoupling coefficients involving fundamental lines [6, 9, 10]. For one pair, the open-chain colour overlaps again depend only on the relative gluon ordering, so the same symmetric-group transform applies once the required open-chain overlaps have been computed. With several pairs, one could apply the transform to gluon permutations within each quark-line arrangement, but different arrangements remain coupled; whether such partial transforms reduce the total cost remains to be determined.

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